sys.internal_tables (Transact-SQL)

Applies to: SQL Server

Returns one row for each object that is an internal table. Internal tables are automatically generated by SQL Server to support various features. For example, when you create a primary XML index, SQL Server automatically creates an internal table to persist the shredded XML document data. Internal tables appear in the `sys` schema of every database and have unique, system-generated names that indicate their function, for example, `xml_index_nodes_2021582240_32001` or `queue_messages_1977058079`.

Expand table

Column name	Data type	Description
Columns inherited from `sys.objects`		For a list of columns that this view inherits, see sys.objects.
`internal_type`	**tinyint**	Type of the internal table: 3 = `QUERY_DISK_STORE_QUERY_HINTS` 4 = `QUERY_DISK_STORE_QUERY_TEMPLATE_PARAMETERIZATION` 6 = `QUERY_DISK_STORE_WAIT_STATS` 201 = `QUEUE_MESSAGES` 202 = `XML_INDEX_NODES` 203 = `FULLTEXT_CATALOG_FREELIST` 204 = `FULLTEXT_INDEX_MAP` 205 = `QUERY_NOTIFICATION` 206 = `SERVICE_BROKER_MAP` 207 = `EXTENDED_INDEXES` (such as a spatial index) 208 = `FILESTREAM_TOMBSTONE` 209 = `CHANGE_TRACKING` 210 = `TRACKED_COMMITTED_TRANSACTIONS` 220 = `CONTAINED_FEATURES` 225 = `FILETABLE_UPDATES` 236 = `SELECTIVE_XML_INDEX_NODE_TABLE` 240 = `QUERY_DISK_STORE_QUERY_TEXT` 241 = `QUERY_DISK_STORE_QUERY` 242 = `QUERY_DISK_STORE_PLAN` 243 = `QUERY_DISK_STORE_RUNTIME_STATS` 244 = `QUERY_DISK_STORE_RUNTIME_STATS_INTERVAL` 245 = `QUERY_CONTEXT_SETTINGS`

Column name	Data type	Description

`internal_type_desc`	**nvarchar(60)**	Description of the type of internal table:
		
		`QUERY_DISK_STORE_QUERY_HINTS`
		`QUERY_DISK_STORE_QUERY_TEMPLATE_PARAMETERIZATION`
		`QUERY_DISK_STORE_WAIT_STATS`
		`QUEUE_MESSAGES`
		`XML_INDEX_NODES`
		`FULLTEXT_CATALOG_FREELIST`
		`FULLTEXT_INDEX_MAP`
		`QUERY_NOTIFICATION`
		`SERVICE_BROKER_MAP`
		`EXTENDED_INDEXES`
		`FILESTREAM_TOMBSTONE`
		`CHANGE_TRACKING`
		`TRACKED_COMMITTED_TRANSACTIONS`
		`CONTAINED_FEATURES`
		`FILETABLE_UPDATES`
		`SELECTIVE_XML_INDEX_NODE_TABLE`
		`QUERY_DISK_STORE_QUERY_TEXT`
		`QUERY_DISK_STORE_QUERY`
		`QUERY_DISK_STORE_PLAN`
		`QUERY_DISK_STORE_RUNTIME_STATS`
		`QUERY_DISK_STORE_RUNTIME_STATS_INTERVAL`
		`QUERY_CONTEXT_SETTINGS`

`parent_id`	**int**	ID of the parent, regardless of whether it's schema-scoped or not. Otherwise, `0` if there's no parent.
		
		`queue_messages` = `object_id` of queue
		
		`xml_index_nodes` = `object_id` of the XML index
		
		`fulltext_catalog_freelist` = `fulltext_catalog_id` of the full-text catalog
		
		`fulltext_index_map` = `object_id` of the full-text index
		
		`query_notification`, or `service_broker_map` = `0`
		
		`extended_indexes` = `object_id` of an extended index, such as a spatial index
		
		`object_id` of the table for which table tracking is enabled = `change_tracking`

Column name	Data type	Description
`parent_minor_id`	**int**	Minor ID of the parent. `xml_index_nodes` = `index_id` of the XML index `extended_indexes` = `index_id` of an extended index, such as a spatial index `0` = `queue_messages`, `fulltext_catalog_freelist`, `fulltext_index_map`, `query_notification`, `service_broker_map`, or `change_tracking`
`lob_data_space_id`	**int**	Non-zero value is the ID of data space (filegroup or partition-scheme) that holds the large object (LOB) data for this table.
`filestream_data_space_id`	**int**	Reserved for future use.

Internal tables don't contain user-accessible data, and their schema are fixed and unalterable. You can't reference internal table names in Transact-SQL statements. For example, you can't execute a statement such as `SELECT * FROM <sys.internal_table_name>`. However, you can query catalog views to see the metadata of internal tables.

Permissions

The visibility of the metadata in catalog views is limited to securables that a user either owns, or on which the user was granted some permission. For more information, see Metadata Visibility Configuration.

Remarks

Internal tables are placed on the same filegroup as the parent entity. You can use the catalog query shown in Example F to return the number of pages internal tables consume for in-row, out-of-row, and large object (LOB) data.

You can use the sp_spaceused system procedure to return space usage data for internal tables. `sp_spaceused` reports internal table space in the following ways:

- When a queue name is specified, the underlying internal table associated with the queue is referenced and its storage consumption is reported.
- Pages that are used by the internal tables of XML indexes, spatial indexes, and full-text indexes are included in the `index_size` column. When a table or indexed view name is

specified, the pages for the XML indexes, spatial indexes, and full-text indexes for that object are included in the columns `reserved` and `index_size`.

Examples

The following examples demonstrate how to query internal table metadata by using catalog views.

A. Show internal tables that inherit columns from the sys.objects catalog view

SQL

```
SELECT * FROM sys.objects WHERE type = 'IT';
```

B. Return all internal table metadata (including that which is inherited from sys.objects)

SQL

```
SELECT * FROM sys.internal_tables;
```

C. Return internal table columns and column data types

SQL

```
SELECT SCHEMA_NAME(itab.schema_id) AS schema_name,
       itab.name AS internal_table_name,
       typ.name AS column_data_type,
       col.*
FROM sys.internal_tables AS itab
     INNER JOIN sys.columns AS col
         ON itab.object_id = col.object_id
     INNER JOIN sys.types AS typ
         ON typ.user_type_id = col.user_type_id
ORDER BY itab.name, col.column_id;
```

D. Return internal table indexes

SQL

```
SELECT SCHEMA_NAME(itab.schema_id) AS schema_name,
       itab.name AS internal_table_name,
```

```
       idx.*
FROM sys.internal_tables AS itab
     INNER JOIN sys.indexes AS idx
         ON itab.object_id = idx.object_id
ORDER BY itab.name, idx.index_id;
```

E. Return internal table statistics

SQL

```
SELECT SCHEMA_NAME(itab.schema_id) AS schema_name,
       itab.name AS internal_table_name,
       s.*
FROM sys.internal_tables AS itab
     INNER JOIN sys.stats AS s
         ON itab.object_id = s.object_id
ORDER BY itab.name, s.stats_id;
```

F. Return internal table partition and allocation unit information

SQL

```
SELECT SCHEMA_NAME(itab.schema_id) AS schema_name,
       itab.name AS internal_table_name,
       idx.name AS heap_or_index_name,
       p.*,
       au.*
FROM sys.internal_tables AS itab
     INNER JOIN sys.indexes AS idx
--     JOIN to the heap or the clustered index
         ON itab.object_id = idx.object_id
        AND idx.index_id IN (0, 1)
     INNER JOIN sys.partitions AS p
         ON p.object_id = idx.object_id
        AND p.index_id = idx.index_id
     INNER JOIN sys.allocation_units AS au
--     IN_ROW_DATA (type 1) and ROW_OVERFLOW_DATA (type 3) => JOIN to partition's
Hobt
--     else LOB_DATA (type 2) => JOIN to the partition ID itself.
         ON au.container_id = CASE au.type WHEN 2 THEN p.partition_id ELSE p.hobt_id
END
ORDER BY itab.name, idx.index_id;
```

G. Return internal table metadata for XML indexes

SQL

```
SELECT t.name AS parent_table,
       t.object_id AS parent_table_id,
```

```
       it.name AS internal_table_name,
       it.object_id AS internal_table_id,
       xi.name AS primary_XML_index_name,
       xi.index_id AS primary_XML_index_id
FROM sys.internal_tables AS it
     INNER JOIN sys.tables AS t
         ON it.parent_id = t.object_id
     INNER JOIN sys.xml_indexes AS xi
         ON it.parent_id = xi.object_id
        AND it.parent_minor_id = xi.index_id
WHERE it.internal_type_desc = 'XML_INDEX_NODES';
GO
```

H. Return internal table metadata for Service Broker queues

SQL

```
SELECT q.name AS queue_name,
       q.object_id AS queue_id,
       it.name AS internal_table_name,
       it.object_id AS internal_table_id
FROM sys.internal_tables AS it
     INNER JOIN sys.service_queues AS q
         ON it.parent_id = q.object_id
WHERE it.internal_type_desc = 'QUEUE_MESSAGES';
GO
```

I. Return internal table metadata for all Service Broker services

SQL

```
SELECT * FROM tempdb.sys.internal_tables
WHERE internal_type_desc = 'SERVICE_BROKER_MAP';
GO
```

Related content

- System catalog views (Transact-SQL)
- Object catalog views (Transact-SQL)

Last updated on 07/08/2025